Applied Econometrics: Session 4

Regression, Dummy Variables, Grouped Data, and Controls

PhD Intensive Course

Lecture 4

April 29, 2026

Why This Lecture Matters

Linear regression is the workhorse of applied causal inference.

But regression has two very different meanings:

- mechanically, it is a best linear summary of conditional means;
- causally, it is valid only when the comparison left inside the coefficient is credible.

Core question

Which variation in treatment remains after we control for other variables, and is that variation as-good-as-random?

Roadmap for Today

1. CEF, OLS, and Causal Regression
2. Omitted-Variable Bias
3. Dummy Variables and Grouped Regression
4. Beyond Confounders
5. Synthesis

Where We Are in the Course

Earlier lectures gave us three ingredients:

- potential outcomes define the causal target;
- random assignment removes selection bias by design;
- regression summarizes conditional averages and partial correlations.

Today we put them together:

Lecture 4 message

Regression adjustment is powerful when it compares treated and untreated units that are comparable after conditioning. It is dangerous when controls create artificial comparisons.

Potential Outcomes Reminder

For each unit i:

$$Y_i = Y_{0i} + T_i(Y_{1i} - Y_{0i}), \qquad T_i \in \{0, 1\}.$$

The average treatment effect is

$$\mathrm{ATE} = \mathbb{E}[Y_{1i} - Y_{0i}].$$

A simple observed difference satisfies

$$\mathbb{E}[Y_i \mid T_i = 1] - \mathbb{E}[Y_i \mid T_i = 0] = \mathrm{ATT} + \{\mathbb{E}[Y_{0i} \mid T_i = 1] - \mathbb{E}[Y_{0i} \mid T_i = 0]\}.$$

- The second term is selection bias.
- Regression is useful only if it helps remove or avoid this term.

Appendix details

Regression with a Treatment Dummy

In a randomized experiment, estimate

$$Y_i = \alpha + \tau T_i + u_i.$$

Since T_i is a dummy:

$$\mathbb{E}[Y_i \mid T_i = 0] = \alpha, \qquad \mathbb{E}[Y_i \mid T_i = 1] = \alpha + \tau.$$

Therefore

$$\tau = \mathbb{E}[Y_i \mid T_i = 1] - \mathbb{E}[Y_i \mid T_i = 0].$$

Interpretation

With a binary treatment and an intercept, OLS is just a difference in means written in regression language.

Appendix details

The Conditional Expectation Function

For outcome Y_i and covariates X_i, define

$$m(x) = \mathbb{E}[Y_i \mid X_i = x].$$

- The CEF is the average outcome among units with the same covariates.
- It is the best unrestricted prediction of Y_i using X_i.
- It can be nonlinear, discontinuous, or defined by many cells.
- It is not automatically causal.

Beginner intuition

If X is "years of education", the CEF asks: among people with exactly x years, what is the average wage?

Appendix details

Population OLS

Regression chooses the best linear approximation:

$$\beta = \arg\min_{b} \mathbb{E}\left[(Y_i - X_i' b)^2\right].$$

The first-order condition gives

$$\mathbb{E}[X_i(Y_i - X_i'\beta)] = 0.$$

If $\mathbb{E}[X_i X_i']$ is invertible,

$$\beta = \mathbb{E}[X_i X_i']^{-1}\mathbb{E}[X_i Y_i].$$

- Orthogonality is a property of the projection.
- It does not say the treatment is randomly assigned.

Appendix details

Regression Anatomy: What Multiple Regression Does

Suppose the regression is

$$Y_i = \alpha + \tau T_i + X_i'\gamma + u_i.$$

Let $\tilde{T}_i$ be the residual from regressing T_i on X_i and a constant. Then the coefficient on treatment is

$$\tau = \frac{\mathrm{Cov}(Y_i, \tilde{T}_i)}{\mathrm{Var}(\tilde{T}_i)}.$$

Intuition

Multiple regression first removes from treatment the part predictable from controls. Then it asks whether the remaining treatment variation predicts the outcome.

Appendix details

When Regression Adjustment Is Causal

Regression adjustment needs a causal assumption, not only algebra:

$$(Y_{0i}, Y_{1i}) \perp\!\!\!\perp T_i \mid X_i.$$

This is conditional independence, or selection on observables.

Under this assumption,

$$\mathbb{E}[Y_i \mid T_i = 1, X_i = x] - \mathbb{E}[Y_i \mid T_i = 0, X_i = x] = \mathbb{E}[Y_{1i} - Y_{0i} \mid X_i = x].$$

Meaning

Within each X-cell, treated and untreated units are comparable in potential outcomes.

Appendix details

Regression Adjustment in Words

A regression with controls tries to imitate a blocked experiment:

1. compare treated and untreated units with the same observed covariates;
2. estimate many within-cell treatment differences;
3. average those differences using weights implied by the regression.

What regression cannot do

It cannot fix unobserved confounders, bad measurement, lack of overlap, or post-treatment conditioning mistakes.

Why Omitting Confounders Biases Regression

A confounder is a variable that affects both treatment and outcome.

$$A_i \to T_i, \qquad A_i \to Y_i.$$

Example: ability in a wage-schooling regression.

- Ability raises wages directly.
- Ability also predicts years of education.
- A short regression of wage on education mixes education and ability.

Plain language

If high-education people would have earned more even without extra education, the education coefficient is too large.

The Omitted-Variable Bias Formula

Suppose the long regression is

$$Y_i = \alpha + \tau T_i + \beta A_i + u_i,$$

but we estimate the short regression Y_i on T_i only. Then

$$\tau_{\text{short}} = \tau + \beta\delta,$$

where

$$\delta = \frac{\mathrm{Cov}(A_i, T_i)}{\mathrm{Var}(T_i)}$$

is the slope from regressing the omitted variable A_i on treatment T_i.

Memorize

Short coefficient = long coefficient + effect of omitted variable on outcome × relationship between omitted and included variable.

Appendix details

Sign Intuition for OVB

Effect of omitted A on Y	Corr. of A with T	Bias direction
Positive	Positive	Upward
Positive	Negative	Downward
Negative	Positive	Downward
Negative	Negative	Upward

Exam habit

State both signs. "The coefficient is biased" is incomplete; the sign depends on both arrows.

OVB Simulation: Education and Wages

The script simulates latent ability and family background.

Specification	Education coefficient	Interpretation
Short: no controls	0.204	badly upward biased
Add noisy proxies	0.143	bias falls but remains
Oracle: true confounders	0.078	close to true 0.080

- The short regression attributes ability and family background to education.
- Imperfect proxies help, but they do not fully reconstruct the experiment.
- The oracle regression is not available in real work; it is a teaching benchmark.

FWL Simulation: What Changed After Controls?

In the same simulation, the education coefficient with proxies is 0.143.

The Frisch–Waugh–Lovell calculation:

1. regress education on test score and parent index;
2. save residualized education $\tilde{Educ}_i$;
3. regress log wage on $\tilde{Educ}_i$.

gives 0.143 again.

Interpretation

The controlled coefficient uses only the part of education not linearly predicted by the included controls.

Controls Are Not a Ritual

Do not add controls because the dataset contains them.

For each proposed control, ask:

1. Is it determined before treatment?
2. Does it help block a back-door path from treatment to outcome?
3. Does it mainly improve outcome prediction without changing treatment assignment?
4. Could treatment itself affect it?
5. Could it be a common effect of treatment and an outcome determinant?

Practical rule

A control list is part of the research design. It needs an argument, not only a software formula.

Binary Dummy Variables

A dummy variable is a 0/1 indicator.

$$Female_i = \begin{cases} 1 & \text{if unit } i \text{ is female,} \\ 0 & \text{otherwise.} \end{cases}$$

In

$$Y_i = \alpha + \beta Female_i + u_i,$$

we have

$$\alpha = \mathbb{E}[Y_i \mid Female_i = 0], \qquad \beta = \mathbb{E}[Y_i \mid Female_i = 1] - \mathbb{E}[Y_i \mid Female_i = 0].$$

Important

A dummy coefficient is a difference relative to a base group.

The Dummy-Variable Trap

Suppose education has categories $8, 9, \ldots, 18$.

If we include an intercept and a dummy for every category, the columns add up to the intercept:

$$1 = D_{8i} + D_{9i} + \cdots + D_{18i}.$$

Then $X'X$ is singular.

Standard solution:

- keep the intercept;
- omit one category as the base group;
- interpret every dummy relative to that omitted category.

Many Dummies: A Saturated Model

With categories $g = 1, \ldots, G$, estimate

$$Y_i = \alpha + \beta_2 D_{2i} + \cdots + \beta_G D_{Gi} + u_i,$$

where group 1 is the base group.

OLS fitted values are group means:

$$\widehat{Y}_i = \begin{cases} \bar{Y}_1 & \text{if } i \in g = 1, \\ \bar{Y}_g & \text{if } i \in g > 1. \end{cases}$$

Interpretation

A saturated dummy regression is a table of conditional means written as a regression.

Appendix details

Dummy Simulation: Flexible Education Effects

Simulated wages have a nonlinear education schedule.

Model	What it imposes	Example output
Linear Y on education	one constant slope	slope $= 1.691$
Education dummies	one mean per year	fitted means equal cell means

Why use dummies?

Dummies avoid forcing a straight line when the CEF is naturally step-shaped or nonlinear.

Treatment Dummies and Interactions

Suppose T_i is treatment and G_i is a group dummy.

$$Y_i = \alpha + \tau T_i + \lambda G_i + u_i.$$

This allows different baseline means by group, but the treatment effect is forced to be the same.

To allow different treatment effects:

$$Y_i = \alpha + \tau T_i + \lambda G_i + \theta(T_i G_i) + u_i.$$

Then

$$\text{effect for } G = 0 : \tau, \qquad \text{effect for } G = 1 : \tau + \theta.$$

Student warning

"Controlling for group" is not the same as allowing the effect to differ by group.

Grouped Data: Why Weights Matter

Sometimes data are aggregated:

$$(\bar{Y}_g, \bar{X}_g, n_g), \qquad g = 1, \ldots, G.$$

If X_i is constant within group g, individual OLS on

$$Y_i = \alpha + \beta X_i + u_i$$

can be recovered from grouped means by weighted least squares:

$$\min_{\alpha,\beta} \sum_g n_g (\bar{Y}_g - \alpha - \beta \bar{X}_g)^2.$$

Reason

A group with 800 observations should count more than a group with 8 observations.

Appendix details

Grouped Regression Simulation

In the simulated education-wage data:

Estimator	Intercept	Education slope
Individual OLS	-2.247	1.691
Grouped WLS with counts	-2.247	1.691
Grouped unweighted OLS	-4.224	1.888

- Weighted grouped regression reproduces the individual-data coefficient here.
- Unweighted grouped regression treats rare and common education levels equally.

When Grouped Regression Is Safe

Grouped regression is safe when:

- the group means correspond to the same units and population;
- the regressors used in the model are available as group means;
- weights are the number of observations behind each mean;
- aggregation does not hide within-group confounding needed for identification.

Common mistake

Regressing totals on totals often changes the question. For grouped OLS, use means with counts as weights.

What Dummies and Groups Teach Us

Dummies and grouped data make a larger point:

- regression can reproduce cell means exactly when the model is saturated;
- parametric regressions impose structure across cells;
- weights decide which cells matter most;
- causal interpretation still depends on comparability within cells.

Bridge to controls

More controls create more cells. This can help identification, but it can also create sparse cells, bad overlap, and bad-control mistakes.

Not Every Control Has the Same Role

Controls can play different roles:

Type	Graphical role	Usual advice
Confounder	$X \to T,\ X \to Y$	include
Outcome predictor	$X \to Y$ only	often include for precision
Treatment predictor	$X \to T$ only	usually avoid if not needed
Mediator	$T \to M \to Y$	exclude for total effect
Collider	$T \to C \leftarrow U \to Y$	exclude

The same word, "control", hides very different causal objects.

Outcome Predictors Can Improve Precision

Suppose treatment is randomized:

$$Y_i = \alpha + \tau T_i + \gamma X_i + \varepsilon_i,$$

where X_i predicts Y_i but not T_i.

Including X_i:

- does not fix bias, because there was no bias from random assignment;
- can reduce residual variance;
- often makes the standard error for $\hat{\tau}$ smaller.

Example

Baseline test scores in an education experiment often improve precision for treatment effects on final test scores.

Treatment Predictors Can Hurt Precision

A variable can predict treatment but add little outcome information after other controls.

By FWL,

$$\hat{\tau} = \frac{\mathrm{Cov}(Y_i, \tilde{T}_i)}{\mathrm{Var}(\tilde{T}_i)}.$$

If a new control predicts T_i, then $\mathrm{Var}(\tilde{T}_i)$ can shrink.

Consequence

Less residual treatment variation means a noisier estimate, unless the control also helps explain the outcome enough to compensate.

Appendix details

Bad Controls: Mediators

A mediator lies on the causal path:

$$T_i \to M_i \to Y_i.$$

If the target is the total effect of T_i on Y_i, do not control for M_i.

Example: email treatment $\to$ customer opens email $\to$ customer pays.

- Total effect asks: what is the effect of sending the email?
- Controlling for opening asks: what is the effect of sending the email holding opening fixed?
- The second question removes part of the mechanism.

Appendix details

Mediator Simulation

Treatment is randomized. It increases agreement, and agreement increases payment.

Regression	Treatment coefficient	Meaning
Payment on treatment	8.636	total effect estimate
Payment on treatment + agreement	2.211	direct effect

Lesson

A control can make a randomized experiment answer the wrong question.

Bad Controls: Colliders and Selection

A collider is a common effect:

$$T_i \to C_i \leftarrow U_i \to Y_i.$$

If we condition on C_i, treatment and the unobserved outcome determinant U_i become associated.

Intuition

Among units with the same C, a high value of treatment must be offset by a lower value of U. This creates a non-causal comparison.

Appendix details

Collider Simulation

Treatment is randomized and the true effect is 6.

Regression	Treatment coefficient	Interpretation
Outcome on treatment	5.999	randomization works
Outcome on treatment + collider	1.700	conditioning creates bias

- The collider is partly caused by treatment.
- It is also partly caused by an unobserved determinant of the outcome.
- Holding it fixed opens a non-causal path from treatment to outcome.

Selection on Outcomes Is Also a Bad Control Problem

Many applied mistakes look like:

$$\text{estimate treatment effect only among units with } Y_i > 0.$$

If treatment affects whether $Y_i > 0$, this selects on a post-treatment outcome.

- Example: marketing campaign affects whether customers buy anything.
- Dropping zero spenders keeps different latent customer types in treatment and control.
- Random assignment before selection does not rescue the selected comparison.

Safer target

Estimate the effect on total spending for everyone, then separately describe participation if that is policy-relevant.

Practical Control-Selection Rules

Before adding X, classify it:

1. **Pre-treatment confounder:** include if it blocks selection bias.
2. **Pre-treatment outcome predictor:** include if it improves precision and does not damage overlap.
3. **Treatment predictor only:** be cautious; it may reduce useful treatment variation.
4. **Mediator:** exclude for total effects; include only for a direct-effect question with extra assumptions.
5. **Collider or selection variable:** exclude.

Workflow

Draw the timing first. If a variable is measured after treatment, treat it as suspicious until proven otherwise.

A Good Regression Table Explains the Design

A regression table should not be a list of increasingly large kitchen-sink specifications.

Better sequence:

1. show raw treatment-control difference;
2. add pre-treatment confounders needed for identification;
3. add precision controls separately if useful;
4. avoid post-treatment variables in the main total-effect specification;
5. explain why the preferred column matches the causal question.

Student habit

Label columns by identification logic, not by "more controls".

What Regression Is Doing

- A treatment dummy without controls is a difference in means.
- OLS approximates the CEF with a linear function.
- Multiple regression uses residual treatment variation after partialling out controls.
- Dummy regressions can reproduce group means exactly.
- Grouped regression needs weights when group sizes differ.

When Regression Is Credible

Regression is credible when:

- the target estimand is clear;
- controls are pre-treatment and justified by the assignment process;
- there is enough overlap after conditioning;
- the functional form is not doing hidden extrapolation;
- post-treatment variables are kept out of total-effect regressions.

One-sentence summary

Regression is not the research design; it is the estimator that implements the design.

Python Examples for This Lecture

The self-contained script is:

`lectures/code/lecture-4-regression-controls.py`

It simulates:

- omitted-variable bias and the FWL theorem;
- dummy variables and grouped weighted regression;
- bad controls through mediators and colliders.

Use in class

Ask students to change the signs and strengths of the simulated relationships, then predict how the coefficients should move before running the code.

Sources and References

Main source chapters:

- Matheus Facure Alves, Chapter 05: The Unreasonable Effectiveness of Linear Regression.
- Matheus Facure Alves, Chapter 06: Grouped and Dummy Regression.
- Matheus Facure Alves, Chapter 07: Beyond Confounders.

Background econometrics:

- Angrist and Pischke, *Mostly Harmless Econometrics*.
- Angrist and Pischke, *Mastering 'Metrics*.
- Hernan and Robins, *Causal Inference: What If*.

What Students Should Be Able to Do

After this lecture, students should be able to:

- explain why a treatment dummy coefficient equals a mean difference;
- use FWL intuition to interpret controlled regressions;
- derive and sign omitted-variable bias;
- interpret dummy coefficients relative to a base group;
- explain why grouped regressions need weights;
- distinguish confounders, precision controls, mediators, and colliders.

Appendix Map

- A1: Selection-bias decomposition reminder.
- A2: Dummy-regression difference in means.
- A3: CEF foundations.
- A4: OLS normal equations.
- A5: Frisch–Waugh–Lovell theorem.
- A6: Conditional independence and regression adjustment.
- A7: Omitted-variable bias formula.
- A8: Dummy regressions and group means.
- A9: Grouped weighted OLS.
- A10: Controls and precision.
- A11: Mediators and total effects.
- A12: Colliders and induced association.

Appendix A1: Selection-Bias Decomposition

Start with

$$\mathbb{E}[Y_i \mid T_i = 1] - \mathbb{E}[Y_i \mid T_i = 0] = \mathbb{E}[Y_{1i} \mid T_i = 1] - \mathbb{E}[Y_{0i} \mid T_i = 0].$$

Add and subtract $\mathbb{E}[Y_{0i} \mid T_i = 1]$:

$$\mathbb{E}[Y_{1i} - Y_{0i} \mid T_i = 1] + \{\mathbb{E}[Y_{0i} \mid T_i = 1] - \mathbb{E}[Y_{0i} \mid T_i = 0]\}.$$

The first term is ATT. The second term is selection bias.

Back to main slide

Appendix A2: Treatment Dummy Equals Difference in Means

Regression:

$$Y_i = \alpha + \tau T_i + u_i, \qquad T_i \in \{0, 1\}.$$

The normal equations imply residuals average to zero within each treatment group:

$$\sum_{T_i=0} (Y_i - \hat{\alpha}) = 0, \qquad \sum_{T_i=1} (Y_i - \hat{\alpha} - \hat{\tau}) = 0.$$

Therefore

$$\hat{\alpha} = \bar{Y}_0, \qquad \hat{\alpha} + \hat{\tau} = \bar{Y}_1,$$

so

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0.$$

Back to main slide

Appendix A3: CEF Foundations

Define

$$m(X_i) = \mathbb{E}[Y_i \mid X_i], \qquad \varepsilon_i = Y_i - m(X_i).$$

Then

$$\mathbb{E}[\varepsilon_i \mid X_i] = \mathbb{E}[Y_i - m(X_i) \mid X_i] = \mathbb{E}[Y_i \mid X_i] - m(X_i) = 0.$$

Hence ε_i is uncorrelated with any function $g(X_i)$:

$$\mathbb{E}[g(X_i)\varepsilon_i] = \mathbb{E}\{g(X_i)\mathbb{E}[\varepsilon_i \mid X_i]\} = 0.$$

Back to main slide

Appendix A4: OLS Normal Equations

Minimize

$$Q(b) = \mathbb{E}[(Y_i - X_i' b)^2].$$

Differentiate:

$$\frac{\partial Q(b)}{\partial b} = -2\mathbb{E}[X_i(Y_i - X_i' b)].$$

At the minimum:

$$\mathbb{E}[X_i(Y_i - X_i'\beta)] = 0.$$

Rearranging:

$$\mathbb{E}[X_i Y_i] = \mathbb{E}[X_i X_i']\beta, \qquad \beta = \mathbb{E}[X_i X_i']^{-1}\mathbb{E}[X_i Y_i].$$

Back to main slide

Appendix A5: Frisch–Waugh–Lovell

Let M_X be the residual-maker after projecting on controls X:

$$M_X = I - X(X'X)^{-1}X'.$$

In

$$Y = T\tau + X\gamma + u,$$

premultiply by M_X:

$$M_X Y = M_X T\tau + M_X X\gamma + M_X u.$$

Since $M_X X = 0$,

$$\tilde{Y} = \tilde{T}\tau + \tilde{u}.$$

Thus

$$\hat{\tau} = (\tilde{T}'\tilde{T})^{-1}\tilde{T}'\tilde{Y}.$$

With a single treatment variable, this is $\mathrm{Cov}(Y, \tilde{T})/\mathrm{Var}(\tilde{T})$.

Back to main slide

Appendix A6: Conditional Independence

If

$$(Y_{0i}, Y_{1i}) \perp\!\!\!\perp T_i \mid X_i,$$

then

$$\mathbb{E}[Y_i \mid T_i = 1, X_i = x] = \mathbb{E}[Y_{1i} \mid T_i = 1, X_i = x] = \mathbb{E}[Y_{1i} \mid X_i = x],$$

and

$$\mathbb{E}[Y_i \mid T_i = 0, X_i = x] = \mathbb{E}[Y_{0i} \mid X_i = x].$$

Subtracting gives

$$\mathbb{E}[Y_i \mid T_i = 1, X_i = x] - \mathbb{E}[Y_i \mid T_i = 0, X_i = x] = \mathbb{E}[Y_{1i} - Y_{0i} \mid X_i = x].$$

Back to main slide

Appendix A7: Omitted-Variable Bias Formula

True long model:

$$Y_i = \alpha + \tau T_i + \beta A_i + u_i, \qquad \mathrm{Cov}(T_i, u_i) = 0.$$

Short-regression slope:

$$\tau_{\text{short}} = \frac{\mathrm{Cov}(Y_i, T_i)}{\mathrm{Var}(T_i)}.$$

Substitute the true model:

$$\tau_{\text{short}} = \frac{\mathrm{Cov}(\alpha + \tau T_i + \beta A_i + u_i, T_i)}{\mathrm{Var}(T_i)}.$$

Using covariance rules:

$$\tau_{\text{short}} = \tau + \beta \frac{\mathrm{Cov}(A_i, T_i)}{\mathrm{Var}(T_i)} = \tau + \beta\delta.$$

Back to main slide

Appendix A8: Dummy Regression Gives Group Means

For groups $g = 1, \ldots, G$, estimate a saturated model with one omitted base group:

$$Y_i = \alpha + \sum_{g=2}^{G} \beta_g D_{gi} + u_i.$$

Fitted values are constant within group. OLS chooses constants c_g to minimize

$$\sum_{g=1}^{G} \sum_{i \in g} (Y_i - c_g)^2.$$

Each c_g solves

$$\frac{\partial}{\partial c_g} \sum_{i \in g} (Y_i - c_g)^2 = -2 \sum_{i \in g} (Y_i - c_g) = 0,$$

so

$$c_g = \bar{Y}_g.$$

Back to main slide

Appendix A9: Grouped Weighted OLS

Individual least squares with groups:

$$\sum_i (Y_i - \alpha - \beta X_i)^2 = \sum_g \sum_{i \in g} (Y_i - \alpha - \beta X_g)^2.$$

Decompose $Y_i = \bar{Y}_g + (Y_i - \bar{Y}_g)$:

$$\sum_{i \in g} (Y_i - \alpha - \beta X_g)^2 = n_g (\bar{Y}_g - \alpha - \beta X_g)^2 + \sum_{i \in g} (Y_i - \bar{Y}_g)^2.$$

The second term does not depend on α, β. Therefore the minimizer solves

$$\min_{\alpha,\beta} \sum_g n_g (\bar{Y}_g - \alpha - \beta X_g)^2.$$

Back to main slide

Appendix A10: Controls and Precision

After residualizing treatment on controls, the treatment standard error has the form

$$\mathrm{se}(\hat{\tau}) \approx \sqrt{\frac{\sigma^2}{\sum_i (\tilde{T}_i - \bar{\tilde{T}})^2}}.$$

Adding a control can affect both terms:

- if it predicts Y, residual variance σ^2 may fall;
- if it predicts T, residual treatment variance may fall;
- the standard error improves only if the first effect dominates the second.

Takeaway

Outcome predictors are often useful precision controls. Pure treatment predictors can make the treatment coefficient noisier.

Back to main slide

Appendix A11: Mediators and Total Effects

Suppose

$$M_i = a + bT_i + v_i, \qquad Y_i = \alpha + \tau T_i + \theta M_i + \varepsilon_i.$$

Substituting the mediator equation into the outcome:

$$Y_i = (\alpha + \theta a) + (\tau + \theta b) T_i + \theta v_i + \varepsilon_i.$$

The total effect is

$$\tau + \theta b.$$

If we control for M_i, the coefficient on T_i is the direct effect τ, not the total effect.

This may be useful for a direct-effect question, but it is wrong for a total-effect policy question.

Back to main slide

Appendix A12: Colliders and Induced Association

Let treatment be randomized:

$$T_i \perp\!\!\!\perp U_i, \qquad Y_i = \alpha + \tau T_i + \lambda U_i + \varepsilon_i.$$

Now control for a common effect

$$C_i = aT_i + bU_i + \eta_i.$$

Before conditioning,

$$\mathrm{Cov}(T_i, U_i) = 0.$$

But after conditioning on C_i, high T_i must be partly offset by low U_i to keep C_i fixed. Thus, generally,

$$\mathrm{Cov}(T_i, U_i \mid C_i) \neq 0.$$

The regression coefficient on T_i then absorbs part of λU_i, even though T_i was randomized.

Back to main slide